LOGISTICS OPERATIONS ANALYTICS

Project Report — Final Submission

Business Analytics Team | Third-Party Logistics (3PL) Company | Nationwide Freight Operations

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1. Executive Summary

This report presents the findings of the Logistics Operations Analytics project, prepared by the Business Analytics Team for the Project Manager. The project responds to concerns raised by the Operations and Finance departments about delivery performance, operating costs, asset utilization and safety risk across the company's nationwide freight network.

Using data from the company's Transportation Management System (TMS) covering January 2022 to December 2024, the team analysed eight performance areas with SQL window functions, common table expressions (CTEs), LAG-based time-series analysis and multi-table joins. The main results are summarised below.

- **Driver Performance:** 124 drivers with recorded trips were compared. Performance is tightly clustered: the fleet on-time rate is 55.68% (range 51.61% to 58.98%, led by DRV00108), average MPG runs from 6.45 to 6.56 (fleet average 6.50) and revenue per mile from $2.09 to $2.21 (led by DRV00101). DRV00001 had the highest MPG in December 2024 (7.00).
- **Route Profitability:** Across 58 routes, RTE00016 earned the highest net profit after fuel ($8.46M) and the highest December 2024 revenue ($353,891.77). Seven routes were loss-making, led by RTE00010 (−$1.42M). Seven routes showed fuel cost rising in each of the last three months, led by RTE00028 (+95%).
- **Fleet Utilization:** Of 92 trucks with trips, TRK00055 recorded the highest mileage (1,417,530) and TRK00040 the lowest (1,178,515) against a fleet average of 1,301,480. TRK00044 generated the most revenue ($3.05M); 44 trucks ran below the average mileage.
- **Maintenance Impact:** TRK00003 had the highest maintenance cost ($90,161.42) and the highest downtime (1,133.1 hours). TRK00014 had the highest cost per mile ($0.0579). TRK00062 had the most downtime in the last six months (329.9 hours).
- **Fuel Efficiency:** Route MPG is nearly uniform (6.47 to 6.54). Fuel cost per mile ranges from $0.39 to $13.69, a spread driven by near-constant fuel cost per route rather than by true efficiency differences, and it needs data validation before use.
- **Customer Revenue:** 200 customers generated $262,525,800.29 over 85,410 loads. The top customer, CUST00200, contributed $1,544,419.81 (0.59%), and the ten largest customers together account for 5.7%, so concentration risk is low.
- **Safety Risk:** DRV00058 recorded the highest incident count (7) and rate (1.06 per 100 trips). Of 170 incidents, 64 (37.6%) were preventable. Annual incidents were 56 (2022), 54 (2023) and 60 (2024).
- **Seasonal Demand:** Demand is steady at about 2,370 loads per month, with annual volumes of 28,589, 28,165 and 28,656. The highest month was August 2022 (2,496 loads), and no recurring seasonal peak was found.

The findings are intended to inform decisions across Operations, Finance, Fleet Management, Sales, Compliance and Executive leadership.

2. Project Overview

The Business Analytics Team was assigned this project as part of an internal operational intelligence initiative. The company, a third-party logistics provider managing nationwide freight operations, required a structured analysis of its Transportation Management System (TMS) data to support decision-making across multiple business functions.

The project mirrors the responsibilities of a data analyst in a 3PL environment: data extraction, multi-table join design, metric derivation, time-series analysis and the formulation of business insights from operational records.

2.1. Scope & Objectives

The scope covers eight analytical modules, each tied to a specific stakeholder group and operational concern:

Module	Stakeholder Group	Core Analytical Question
Driver Performance	Operations Leadership	Which drivers are performing at or above fleet benchmarks?
Route Profitability	Finance Department	Which freight lanes generate profit and which are cost overruns?
Fleet Utilization	Fleet Management	Are company-owned trucks being efficiently deployed?
Maintenance Impact	Maintenance & Operations	What is the financial and operational cost of truck downtime?
Fuel Efficiency	Transportation Planning	Which routes exhibit inefficient fuel consumption?
Customer Revenue	Sales Leadership	Which customers contribute the greatest revenue value?
Safety Risk	Compliance & Safety	Where are incident and preventable-incident rates highest?
Seasonal Demand	Strategic Planning	How does demand vary by month, quarter and season?

3. Methodology & Data Notes

The following definitions and conventions apply throughout the report. They explain how each metric was built and how the figures should be read.

3.1. Metric Definitions

- **Revenue** is the revenue recorded on each load. **Fuel cost** is the total cost of fuel purchases recorded against trips. **Net route profitability** is revenue minus fuel cost only; it excludes driver pay, maintenance, equipment and overhead, so it is a margin after fuel and not a full profit figure.
- **On-time rate** is calculated from delivery event records (on-time events divided by all delivery events), which hold more than one event per trip. **MPG** is miles driven divided by gallons purchased. **Revenue per mile** is load revenue divided by miles driven.
- **Incident rate** is incidents per 100 trips. A **preventable** incident is one flagged as preventable in the safety records.
- The observation period is January 2022 to December 2024. “Most recent month” means December 2024, and “this year” means 2024, the latest year in the data.

3.2. Reading the Results

- Metrics were aggregated separately at each level (driver, route, truck, customer, month) before being joined, to avoid double counting when trips link to several fuel, delivery or incident records.
- Customers are identified by customer ID. Several customer names are shared by more than one ID, so names alone are not unique.
- Distance and per-mile figures are reported as recorded in the TMS data.
- Rankings using DENSE_RANK produce tiers with ties and no gaps. Several metrics (on-time rate, MPG, revenue per mile) vary within narrow bands, so rank differences are small in absolute terms.
- Incident counts are small. A single additional incident moves a driver’s rate noticeably, so rates are best read together with counts.
- Month-level results are affected by calendar length. February has fewer days and is consistently the lowest month for volume and revenue; this is a calendar effect more than a change in demand.

4. Data Architecture & Sources

All analysis was performed against the company's TMS relational database, which holds 14 tables across master, transactional and operational data. The tables used in this project and the join paths between them are shown below.

Table	Domain	Primary Key	Key Fields	Joined To
trips	Transportation	trip_id	load_id, driver_id, truck_id, trailer_id, distance, fuel use	loads, delivery_events, fuel_purchases, safety_incidents
loads	Freight	load_id	customer_id, route_id, load_date, revenue	trips, customers, routes
delivery_events	Delivery	event_id	trip_id, load_id, on-time status	trips
fuel_purchases	Fuel Management	fuel_purchase_id	trip_id, truck_id, driver_id, cost, gallons	trips
maintenance_records	Maintenance	maintenance_id	truck_id, cost, downtime hours, date	trucks (and trips via truck_id)
safety_incidents	Safety	incident_id	trip_id, truck_id, driver_id, preventable flag, incident date	trips
customers	CRM	customer_id	customer name and account details	loads
routes, drivers, trucks	Master data	route_id, driver_id, truck_id	route, driver and equipment attributes	loads, trips, maintenance_records

Joins follow the schema's key relationships: trips to loads is one-to-one; loads to customers and to routes are many-to-one; and fuel purchases, delivery events and safety incidents link many-to-one to trips. Aggregations were performed before joining wherever a one-to-many join could inflate totals.

5. Analytical Framework

The project used a four-pillar SQL analytics framework, selected to match the complexity of each business question:

Technique	Description	Modules Applied
Window Functions	RANK(), DENSE_RANK(), NTILE(), AVG() OVER and rolling frames (ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) to rank, compare and smooth metrics within peer groups without collapsing result sets.	Modules 1 to 8
Common Table Expressions (CTEs)	Break multi-step logic into readable sub-queries for fleet-average and median benchmarks, share-of-total calculations and filtered aggregations.	Modules 1, 2, 3, 5, 6, 7, 8
LAG / Time-Series Analysis	LAG() with monthly or quarterly date truncation to measure month-over-month change, growth rates and directional trends.	Modules 1, 2, 3, 6, 7, 8
Multi-Table Joins	INNER and LEFT joins across two to four tables to build cross-domain metrics such as maintenance cost per mile and incident rate per 100 trips.	Modules 1 to 8

MOD 01 DRIVER PERFORMANCE INSIGHTS

Business Requirement

Operations leadership required an evaluation of driver-level performance to identify high-performing drivers and those linked to delivery delays or operational inefficiency. Metrics were needed for each driver and relative to fleet benchmarks.

Key Metrics Derived

- On-Time Delivery Rate per Driver
- Average Fuel Efficiency (MPG) per Driver
- Revenue Generated per Mile Driven per Driver
- Month-over-Month Revenue Trend per Driver

Analytical Techniques

- RANK() OVER (PARTITION BY month ORDER BY on_time_rate DESC): monthly on-time performance ranking
- NTILE(10): identification of the top 10% of drivers by revenue per mile
- CTE + AVG() OVER (): driver MPG compared with fleet average
- LAG() OVER (PARTITION BY driver_id ORDER BY month): month-over-month revenue trend

Operational Findings

On-Time Delivery Performance

55.68%	58.98%	51.61%
Fleet On-Time Delivery Rate	Highest On-Time Rate — DRV00108	Lowest On-Time Rate — DRV00133
93,196 of 167,392 delivery events	*+3.30 pts above fleet*	*−4.07 pts below fleet*

On-time delivery was measured from delivery event records for all 124 drivers (167,392 events in total, 1,208 to 1,498 per driver). The **fleet-wide on-time rate is 55.68%**, which means about 44% of delivery events are not recorded as on time. **DRV00108** is the best performer at **58.98%** (867 of 1,470 events), 3.30 points above the fleet rate, followed by DRV00038 (58.58%), DRV00138 (58.27%), DRV00112 (58.16%) and DRV00020 (58.14%).

At the other end, **DRV00133** (51.61%) and **DRV00074** (51.82%) trail the fleet by 3.9 to 4.1 points and are well separated from the next-lowest drivers, DRV00122 (53.06%), DRV00124 (53.49%) and DRV00064 (53.52%). Across the fleet, 59 drivers are above the fleet rate and 65 below. The full range is only 7.4 points (51.61% to 58.98%), so most drivers sit within about two points of the fleet rate and the shortfall in delivery reliability is systemic and not limited to a few individuals. Monthly rankings (window function) allow managers to track rank movement and intervene when a driver's position deteriorates.

Position	Driver	Delivery Events	On-Time Events	On-Time Rate	vs Fleet
1	DRV00108	1,470	867	58.98%	+3.30
2	DRV00038	1,410	826	58.58%	+2.90
3	DRV00138	1,366	796	58.27%	+2.59
4	DRV00112	1,262	734	58.16%	+2.48
5	DRV00020	1,388	807	58.14%	+2.46
120	DRV00064	1,220	653	53.52%	−2.16
121	DRV00124	1,404	751	53.49%	−2.19
122	DRV00122	1,208	641	53.06%	−2.62
123	DRV00074	1,264	655	51.82%	−3.86
124	DRV00133	1,366	705	51.61%	−4.07

Revenue per Mile Performance

$2.21
Highest Revenue per Mile
Driver DRV00101

$2.09 – $2.21
Range Across 124 Drivers
average about $2.15

DRV00101 recorded the highest revenue per mile at $2.21, ahead of DRV00002 ($2.20) and DRV00005, DRV00033 and DRV00016 (each $2.19). The lowest values are DRV00012 ($2.09) and DRV00045 and DRV00071 ($2.10). NTILE(10) splits the 124 drivers into groups of 12 or 13, and the top group holds 13 drivers, from $2.17 to $2.21. The 13th and 14th drivers tie at $2.17, so the cut-off falls within a tie. The overall spread of roughly 6% shows that revenue yield per mile is consistent across the driver base.

Rank	Driver	Miles Driven	Revenue	Revenue per Mile
1	DRV00101	952,807	$2,101,700.26	$2.21
2	DRV00002	943,246	$2,073,621.50	$2.20
3	DRV00005	923,128	$2,022,357.46	$2.19
4	DRV00033	958,305	$2,094,122.41	$2.19
5	DRV00016	1,042,631	$2,278,574.88	$2.19
6	DRV00056	979,155	$2,136,656.62	$2.18
7	DRV00057	911,374	$1,984,192.01	$2.18
8	DRV00144	992,420	$2,162,810.15	$2.18
9	DRV00091	855,939	$1,867,304.47	$2.18
10	DRV00130	973,944	$2,118,373.98	$2.18

Fuel Efficiency (MPG) Analysis

Fuel efficiency was benchmarked against the fleet average using a CTE and a window average. The fleet average is **6.50 MPG**, and individual driver averages fall between **6.45 and 6.56 MPG**, a spread of about 1.7%. Differences between drivers are therefore marginal, and fuel efficiency alone does not separate strong drivers from weak ones. Drivers above 6.50 are only slightly more efficient than the fleet norm.

In the most recent month (December 2024), **DRV00001** recorded the highest monthly average MPG at **7.00**, about 0.5 MPG above the fleet average. This is a single-month result and well above the full-period driver averages (6.45 to 6.56), so it is worth studying for driving practices that could be replicated, but it should not be read as DRV00001's typical performance.

> **Key Question — Resolved:** *Which driver had the highest MPG in the most recent month? DRV00001 recorded the highest average MPG of 7.00 in December 2024, found by averaging MPG per driver per month and filtering to the latest month.*

Month-over-Month Revenue Trend

Driver-level monthly revenue was tracked with LAG() to show the change from each driver's previous month. These series move up and down month to month without a sustained direction, so month-over-month swings should be read against a driver's own trend and not as performance signals on their own.

Result Notes

- Delivery records hold about two events per trip, so on-time rates reflect all delivery events.
- NTILE splits drivers into equal-sized groups, so drivers with tied values can fall on either side of the top-10% cut-off.
- Data runs to December 2024.

Data Sources

Tables: trips, delivery_events, loads

Join Path: trips → delivery_events (trip_id) | trips → loads (load_id)

MOD 02 ROUTE PROFITABILITY ASSESSMENT

Business Requirement

Finance stakeholders required visibility into which freight lanes generate profit and which are cost overruns driven by fuel expenditure, to support pricing reviews, route rationalisation and cost management.

Key Metrics Derived

- Total Revenue per Route
- Total Fuel Cost per Route
- Net Route Profitability (Revenue minus Fuel Cost)
- Monthly Profit Trend per Route

Analytical Techniques

- Time series: monthly profit trend using date-based GROUP BY aggregation
- DENSE_RANK() OVER (PARTITION BY quarter ORDER BY net_profit DESC): top 5 routes per quarter
- LAG() OVER (PARTITION BY route_id ORDER BY month): three-month fuel cost escalation detection
- CTE: route profitability compared with the company average

Operational Findings

Route Profit Distribution

58	$8.46M	$2.88M
Routes Analysed	Highest Net Profit *Route RTE00016*	Average Route Profit *company benchmark*

Net profitability after fuel was compared across all 58 routes. **RTE00016** earned the highest net profit at **$8.46 million**, followed by RTE00044, RTE00012, RTE00048, RTE00029, RTE00046 and RTE00042. The company-average route profit is **$2,878,151.87**; 27 routes sit above it and 31 below, so profit is not evenly spread, with a smaller group of strong lanes lifting the average.

Loss-Making Routes

Seven routes earned less than their fuel cost: **RTE00010** (−$1.42M, the largest loss), RTE00011, RTE00015, RTE00022, RTE00033, RTE00036 and RTE00047. These lanes should be reviewed for rate levels, load density and distance before further volume is assigned to them.

Fuel Cost and Revenue

Fuel cost per route sits in a narrow band of roughly **$1.55M to $1.79M** regardless of route, while route revenue and profit vary widely. In practice, a route's profitability is determined mainly by the revenue it earns, not by fuel spend. This uniform fuel cost also explains the unusual fuel-cost-per-mile values reported in Module 5.

Highest Revenue in the Most Recent Month

$353,891.77
Highest December 2024 Revenue
Route RTE00016

Rising Fuel Cost, October to December 2024

A LAG analysis flagged seven routes where fuel cost rose in each of the last three months:

Route	Oct 2024	Nov 2024	Dec 2024	Change Oct to Dec	% Change
RTE00028	$29,907.11	$41,450.10	$58,365.87	+$28,458.76	+95.2%
RTE00055	$40,123.30	$43,550.31	$52,342.86	+$12,219.56	+30.5%
RTE00046	$39,342.92	$42,652.58	$50,090.81	+$10,747.89	+27.3%
RTE00017	$38,216.93	$41,022.23	$44,793.14	+$6,576.21	+17.2%
RTE00006	$34,592.94	$36,678.16	$44,462.05	+$9,869.11	+28.5%
RTE00038	$33,470.33	$37,122.54	$43,314.96	+$9,844.63	+29.4%
RTE00023	$25,663.04	$32,360.46	$35,471.92	+$9,808.88	+38.2%
Total	**$241,316.57**	**$274,836.38**	**$328,841.61**	**+$87,525.04**	**+36.3%**

RTE00028 stands out: its fuel cost nearly doubled in two months (+95.2%). RTE00023 (+38.2%) and RTE00055 (+30.5%) follow. These lanes should be investigated for load changes, driver fuelling behaviour or fuel price changes before the trend carries into 2025. Only three months of data are used, so the trend should be confirmed against later periods.

> **Key Question — Resolved:** *Which route generated the highest revenue last month? Route RTE00016 recorded the highest monthly revenue of $353,891.77 in December 2024, found by filtering loads to the most recent month and ranking routes by SUM(revenue) DESC.*

Data Sources

Tables: loads, trips, fuel_purchases

Join Path: loads → trips (load_id) | trips → fuel_purchases (trip_id)

MOD 03 FLEET UTILIZATION MONITORING

Business Requirement

Fleet management required an assessment of whether company-owned trucks are deployed efficiently and contribute proportionally to revenue, to support asset reallocation, retirement planning and utilisation benchmarking.

Key Metrics Derived

- Total Miles Driven per Truck
- Total Revenue Generated per Truck
- Monthly Revenue per Truck
- Rolling 3-Month Utilisation per Truck

Analytical Techniques

- SUM() OVER (PARTITION BY truck_id ROWS BETWEEN 2 PRECEDING AND CURRENT ROW): rolling 3-month mileage
- RANK() OVER (PARTITION BY month ORDER BY revenue DESC): revenue rank within each month
- CTE: identification of trucks operating below fleet average mileage
- LAG() OVER (PARTITION BY truck_id ORDER BY month): monthly utilisation growth rate

Operational Findings

Distance Utilisation

1,417,530	1,301,480	1,178,515
Highest Mileage	Fleet Average Mileage	Lowest Mileage
Truck TRK00055	*92 trucks with trips*	*Truck TRK00040*

Total miles were computed for the 92 trucks with trips. **TRK00055** drove the most (1,417,530 miles). **TRK00040** drove the least (1,178,515 miles), about 122,965 miles (9.4%) below the fleet average of 1,301,480.28. In total, **44 of 92 trucks** operated below the average. The gap between the highest and lowest truck equals about 18% of the fleet average, which indicates a fairly even deployment of the fleet, with TRK00040 the clearest candidate for reassignment.

Revenue Generation by Truck

$3.05M	$2.53M
Highest Revenue Truck	Lowest Revenue Truck
TRK00044	*TRK00079*

Revenue was attributed to trucks by joining trips to their loads. **TRK00044** generated the most at about $3.05 million; **TRK00079** the least at about $2.53 million. The gap of roughly 20% shows that truck revenue depends on the loads each truck is given as well as on its miles.

Monthly Utilisation Patterns

Monthly revenue per truck, the rolling 3-month mileage and the month-over-month growth rate all fluctuate from month to month without a lasting direction for most trucks. The monthly revenue rank also changes often, so a truck's position in one month says little about the next. Rolling averages are the better basis for utilisation reviews than single-month figures.

Result Notes

- Mileage totals are as recorded in the TMS data and are reported without adjustment.
- The fleet average is based on trucks with at least one trip.

Data Sources

Tables: trips, loads

Join Path: trips → loads (load_id)

MOD 04 MAINTENANCE COST IMPACT ANALYSIS

Business Requirement

The Maintenance and Operations teams required a quantitative assessment of the financial and operational impact of truck maintenance, to find high-cost and high-downtime assets and to understand how maintenance relates to trip volume and revenue.

Key Metrics Derived

- Total Maintenance Cost per Truck
- Total Downtime Hours per Truck
- Maintenance Cost per Mile Driven
- Downtime Impact on Monthly Trip Count

Analytical Techniques

- Time series: maintenance cost trend over time using aggregation
- Window ranking: top 5 trucks by downtime in the last six months
- CTE + join: downtime hours joined to monthly trip counts
- Join: maintenance cost compared with revenue per truck

Operational Findings

Highest Maintenance Cost and Downtime

$90,161.42	1,133.1 hrs	$0.0579/mi
Highest Maintenance Cost	Highest Total Downtime	Highest Cost per Mile
Truck TRK00003	*Truck TRK00003*	*Truck TRK00014*

Maintenance records cover **109 trucks**. **TRK00003** has both the highest maintenance cost ($90,161.42) and the highest downtime (1,133.1 hours), so cost and lost availability are concentrated on one asset. It should be assessed for overhaul or replacement. The lowest-cost truck is TRK00083 ($11,896.28), which also has the lowest cost per mile ($0.0093).

On a per-mile basis, **TRK00014** is highest at $0.0579 per mile, about six times the lowest, which marks it as the truck whose upkeep weighs most on each load it hauls. Maintenance cost is small relative to the revenue each truck earns, at roughly 0.4% to 2.7% of revenue per truck.

Recent Downtime, Last Six Months

A window ranking over the six months to December 2024 identified the five trucks with the most recent downtime:

329.9 hrs
Highest Recent Downtime (6 months)
Truck TRK00062

Rank	Truck	Downtime (Hours)
1	TRK00062	329.90
2	TRK00038	243.10
3	TRK00105	234.00
4	TRK00089	226.40
5	TRK00120	223.50

TRK00062 stands out, with roughly 35% more recent downtime than the next truck. It should be placed on a priority maintenance review.

Downtime and Trip Volume

Downtime hours were joined to monthly trip counts to test whether maintenance events reduce trips. The comparison shows **no clear relationship**: trucks with high downtime do not consistently complete fewer trips, so downtime shows no visible effect on throughput in this data.

> **Key Question — Resolved:** *Which truck had zero downtime last quarter? Using a LEFT JOIN between trucks and maintenance records for the final quarter of 2024, 18 trucks recorded zero or no downtime hours.*

Data Sources

Tables: maintenance_records, trips

Join Path: maintenance_records → trips (truck_id)

MOD 05 FUEL EFFICIENCY ANALYSIS

Business Requirement

The transportation planning team required an analysis of fuel consumption patterns at route level to find inefficient lanes and inform re-routing, driver training and fuel cost management.

Key Metrics Derived

- Average MPG per Route
- Fuel Cost per Mile per Route
- MPG Trend per Route by Month
- Rolling Average MPG per Route

Analytical Techniques

- Time series: monthly MPG trend per route using dispatch date aggregation
- RANK / DENSE_RANK() OVER (ORDER BY avg_mpg DESC): route-level fuel efficiency ranking
- CTE: identification of routes with MPG below the fleet median
- AVG() OVER (PARTITION BY route_id ROWS BETWEEN 2 PRECEDING AND CURRENT ROW): rolling average MPG

Operational Findings

Route-Level Fuel Efficiency

6.47 – 6.54	6.5010
Average MPG Range Across 58 Routes	Fleet Median MPG *29 routes below*

Average MPG per route sits in a very narrow band from **6.47 to 6.54**, a gap of about 1%. The ranking therefore produces only eight distinct DENSE_RANK values, with many routes tied. **29 routes** fall below the fleet median of 6.5010, but the shortfall is small for each, so no lane is a clear fuel-efficiency problem on MPG alone.

Fuel Cost per Mile

$0.39	$13.69
Lowest Fuel Cost per Mile *Route RTE00025*	Highest Fuel Cost per Mile *Route RTE00015*

Fuel cost per mile ranges from **$0.39** (RTE00025) to **$13.69** (RTE00015), a 35-fold difference. This spread is out of line with the near-identical MPG across routes. It arises because fuel cost per route is almost constant (about $1.55M to $1.79M) while route mileage differs widely. Fuel cost per mile should therefore be read as a data-quality flag. It needs validation of how fuel purchases are attributed to routes and trips before it supports pricing or re-routing decisions.

Monthly and Rolling MPG Trends

Monthly MPG per route is noisy, and the 3-month rolling average smooths most of this out. **RTE00021** is the only route showing a sustained decline in rolling MPG and is the best candidate for a closer look at vehicle assignment and driving conditions.

Data Sources

Tables: trips, loads, fuel_purchases

Join Path: loads → trips (load_id) | trips → fuel_purchases (trip_id)

MOD 06 CUSTOMER REVENUE ANALYSIS

Business Requirement

Sales leadership required identification of the highest-value customers and an assessment of customer-level revenue contribution relative to the total portfolio, to support account prioritisation, upsell targeting and concentration-risk evaluation.

Key Metrics Derived

- Total Revenue per Customer
- Average Revenue per Load per Customer
- Monthly Revenue Trend per Customer
- Customer Revenue Share vs Total Company Revenue

Analytical Techniques

- Time series: monthly revenue trend per customer using load date aggregation
- DENSE_RANK() OVER (PARTITION BY quarter ORDER BY revenue DESC): top customers by quarter
- CTE: customer revenue as a percentage of total, using SUM() OVER ()
- LAG() OVER (PARTITION BY customer_id ORDER BY month): month-over-month revenue growth rate

Operational Findings

Revenue by Customer

200	$262.5M	$1.31M
Customers	Total Revenue *85,410 loads*	Average per Customer *median $1.31M*

Total revenue was aggregated for all 200 customers. **CUST00200** is the largest at $1,544,419.81, closely followed by CUST00181 ($1,542,321.02) and CUST00077 ($1,521,982.07). The smallest is **CUST00149** at $1,078,045.01. Revenue is very evenly spread: the mean is $1,312,629, the median $1,305,217, each customer contributes between 0.41% and 0.59% of the total, and the **top ten customers together account for only 5.7%**. Concentration risk is low, but no single account carries much negotiating leverage either. The five largest and five smallest customers are shown below.

Rank	Customer ID	Total Revenue	Share of Total
1	CUST00200	$1,544,419.81	0.59%
2	CUST00181	$1,542,321.02	0.59%
3	CUST00077	$1,521,982.07	0.58%
4	CUST00097	$1,487,129.31	0.57%
5	CUST00122	$1,483,188.90	0.56%
196	CUST00109	$1,152,769.50	0.44%
197	CUST00151	$1,148,596.86	0.44%
198	CUST00071	$1,142,845.27	0.44%
199	CUST00156	$1,128,763.05	0.43%
200	CUST00149	$1,078,045.01	0.41%

Revenue per Load

$3,294.30	$3,073.71	$2,695.11
Highest Revenue per Load *CUST00070*	Company Average per Load	Lowest Revenue per Load *CUST00149*

Customer ID	Loads	Total Revenue	Revenue per Load
CUST00070	430	$1,416,550.21	$3,294.30
CUST00193	415	$1,360,973.63	$3,279.45
CUST00081	433	$1,417,040.84	$3,272.61
CUST00040	397	$1,292,863.72	$3,256.58
CUST00027	417	$1,357,303.31	$3,254.92
CUST00114	400	$1,160,827.89	$2,902.07
CUST00055	429	$1,244,544.47	$2,901.04
CUST00148	428	$1,236,375.06	$2,888.73
CUST00025	413	$1,181,340.71	$2,860.39
CUST00149	400	$1,078,045.01	$2,695.11

Average revenue per load ranges from $2,695.11 to $3,294.30. **CUST00070** earns the most per load and **CUST00149** the least, about 12% below the company average, and it is also last in total revenue. The five lowest-yield customers (CUST00149, CUST00025, CUST00148, CUST00055 and CUST00114) are candidates for a rate review. The top-five table shows that high revenue per load does not mean a high total: CUST00070 ranks below CUST00200 on total revenue because it ships fewer loads (430 against 476).

Monthly Trend and Growth

Monthly revenue per customer is volatile and mean-reverting, with no sustained upward or downward trend for customers as a group. The quarterly number-one customer rotates across 12 different customers, so leadership in any given quarter is not persistent. The largest month-over-month growth rates are low-base effects, such as CUST00056 in July 2024 (+847.56%), CUST00151 in January 2024 (+763.31%) and CUST00076 in May 2022 (+701.24%). Sharp drops occur as well (for example CUST00120 in October 2022,

−90.96%). These swings reflect shipment timing for individual accounts and should be read as normal variation.

> **Key Question — Resolved:** *Which customer contributed the most revenue this year? For 2024, the latest year in the data, CUST00033 contributed the most at $538,785.58 (rank 1, no tie), found by filtering loads to 2024 and ranking customers by SUM(revenue) DESC. Customer names are not unique, so IDs are used.*

Data Sources

Tables: customers, loads

Join Path: customers → loads (customer_id)

MOD 07 SAFETY RISK ASSESSMENT

Business Requirement

The Compliance and Safety departments required a data-driven assessment of incident patterns across the driver population, with a focus on preventable incident rates and trends over time, to support targeted intervention and regulatory compliance.

Key Metrics Derived

- Incident Count per Driver
- Preventable Incident Rate per Driver
- Monthly and Annual Incident Trend
- Incident Rate per 100 Trips

Analytical Techniques

- Time series: monthly incident trend using incident date aggregation with LAG()
- DENSE_RANK() OVER (ORDER BY incident_count DESC): driver ranking by incident frequency
- CTE: preventable vs non-preventable incident ratio
- Join and CTE: incident counts combined with trip counts for the incident rate per 100 trips

Operational Findings

Driver Incident Analysis

7 incidents	1.06 per 100 trips	92 of 124
Highest Incident Count	Highest Incident Rate	Drivers with at Least One Incident
Driver DRV00058	*DRV00058 (660 trips)*	*32 had none*

DRV00058 stands out with 7 incidents, a rate of 1.06 per 100 trips (660 trips). **DRV00003** is next with 6 (0.89 per 100 trips). Both run at roughly twice the rate of anyone else.

Incidents by driver fall into five tiers (DENSE_RANK):

Rank	Incidents per Driver	Number of Drivers	Drivers
1	7	1	DRV00058
2	6	1	DRV00003
3	3	16	DRV00006, 12, 18, 47, 50, 53, 66, 70, 98, 100, 101, 104, 119, 121, 132, 144
4	2	34	Various
5	1	40	Various

Most drivers with an incident had only one or two. The 18 drivers with three or more account for about 36% of driver-linked incidents, so a focused intervention makes more sense than a fleet-wide one. Because 90 of the 92 drivers share ranks 3 to 5, a finer ranking would need a tiebreaker such as severity.

Incident Rate per 100 Trips

Drivers made similar numbers of trips (604 to 749 each), so adjusting for trips barely changes the order. The highest rates:

Rank	Driver	Incidents	Trips	Incidents per 100 Trips
1	DRV00058	7	660	1.06
2	DRV00003	6	677	0.89
3	DRV00012	3	608	0.49
4	DRV00104	3	640	0.47
5	DRV00006	3	647	0.46
6	DRV00018	3	659	0.46
7	DRV00050	3	658	0.46
8	DRV00053	3	651	0.46

Every driver with three incidents rates higher (0.41 to 0.49) than any driver with two (0.27 to 0.33), who in turn rate above those with one (0.14 to 0.16). The 32 drivers with none each completed 607 to 729 trips. DRV00101, the top revenue-per-mile driver in Module 1, has three incidents (0.45 per 100 trips), so productivity and safety don't always go together.

Preventable Incident Rate per Driver

Of the 92 drivers with incidents, 48 had at least one preventable incident and 44 had none. Ranking by rate alone puts one- and two-incident drivers on top (18 drivers sit at 100%), so rates need to be read with counts. The drivers with the most preventable incidents:

Driver	Preventable	Total Incidents	Preventable Rate
DRV00070	3	3	100.00%
DRV00098	3	3	100.00%
DRV00104	3	3	100.00%
DRV00058	3	7	42.86%
DRV00101	2	3	66.67%
DRV00003	2	6	33.33%

DRV00070, DRV00098 and DRV00104 had all three of their incidents classed as preventable, which makes them the strongest candidates for coaching (DRV00013 and DRV00033 were 2 for 2). DRV00058 has the most incidents overall and three preventable ones. DRV00003 has 6 incidents but only 2 preventable, so most of its incidents were outside the driver's control.

Preventable vs Non-Preventable Breakdown

Category	Incident Count	Share of Total	Implication
Preventable	64	37.6%	Attributable to controllable operational factors
Non-Preventable	106	62.4%	External or unavoidable events
Total	**170**	**100%**	**Ratio 0.60 preventable per non-preventable**

For every non-preventable incident there are about 0.6 preventable ones. Roughly three in eight incidents could in principle be reduced through driver behaviour programmes and training.

Monthly and Annual Incident Trend

Incidents came to 56 in 2022, 54 in 2023 (−2, −3.6%) and 60 in 2024 (+6, +11.1%). At 54 to 60 a year, or about 4.7 a month, a swing of that size can easily happen by chance, so the 2024 rise is something to watch and not a confirmed deterioration. Counts by month:

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
2022	7	2	3	4	3	7	6	9	5	4	4	2	56
2023	8	3	5	6	7	1	7	3	3	4	3	4	54
2024	6	3	4	5	7	4	4	6	6	4	9	2	60

Monthly counts range from 1 (June 2023) to 9 (August 2022 and November 2024). Spikes tend to be followed by a drop (9 in November 2024, then 2 in December), which looks like random variation, and no seasonal pattern shows up.

Data Sources

Tables: safety_incidents, trips

Join Path: safety_incidents → trips (trip_id)

MOD 08 SEASONAL DEMAND PATTERN ANALYSIS

Business Requirement

The strategic planning team required identification of seasonality in shipment volume and revenue to support demand forecasting, driver scheduling and capacity allocation.

Key Metrics Derived

- Monthly and Annual Load Volume
- Monthly Revenue Trend
- Monthly Trip Volume
- Quarter-wise Shipment Trend

Analytical Techniques

- LAG() OVER (ORDER BY month): month-over-month load volume growth
- DENSE_RANK() OVER (ORDER BY load_volume DESC): peak month identification
- CTE: revenue contribution by season
- Time series: quarter-wise aggregation using load date and dispatch date

Operational Findings

Annual and Monthly Volume

Year	Loads / Trips	Year-over-Year Change	Revenue	Assessment
2022	28,589	—	$87,850,899.99	Baseline year
2023	28,165	−424 (−1.5%)	$86,955,549.64	Marginal decline
2024	28,656	+491 (+1.7%)	$87,719,350.66	Recovery above 2022
Total	**85,410**		**$262,525,800.29**	

Demand is steady, with 28,165 to 28,656 loads a year. The monthly average is about 2,373 loads, and every month lands between 10% below and 5% above that (2,136 to 2,496). Trip counts by dispatch month match load counts exactly, confirming that trips and loads are one-to-one.

Monthly revenue tracks load volume closely, averaging about $7.29 million (from $6.57M in February 2022 to $7.72M in August 2022). **February is the lowest month every year**, mostly because it is shorter: per day it runs at about 78 loads, in line with other months. March then rebounds each year (+191, +276 and +164 loads). Monthly loads and revenue are below.

Month	Loads 2022	Loads 2023	Loads 2024	Revenue 2022	Revenue 2023	Revenue 2024
Jan	2,466	2,437	2,401	$7,566,362	$7,428,081	$7,386,201
Feb	2,179	2,136	2,267	$6,574,259	$6,618,783	$6,970,221
Mar	2,370	2,412	2,431	$7,401,197	$7,509,746	$7,453,599
Apr	2,346	2,336	2,354	$7,156,041	$7,289,831	$7,122,949
May	2,428	2,348	2,442	$7,518,061	$7,243,795	$7,569,587

Month	Loads 2022	Loads 2023	Loads 2024	Revenue 2022	Revenue 2023	Revenue 2024
Jun	2,376	2,312	2,418	$7,325,638	$7,278,443	$7,489,030
Jul	2,421	2,433	2,394	$7,380,097	$7,576,414	$7,310,742
Aug	2,496	2,344	2,426	$7,716,840	$7,194,394	$7,349,694
Sep	2,304	2,327	2,339	$7,108,890	$7,018,436	$7,145,569
Oct	2,468	2,397	2,474	$7,501,832	$7,406,524	$7,497,106
Nov	2,360	2,364	2,357	$7,260,206	$7,235,373	$7,095,886
Dec	2,375	2,319	2,353	$7,341,477	$7,155,729	$7,328,766

Peak Shipment Months

Year	Peak Month	Loads	Observation
2022	August	2,496	Highest month in the dataset
2023	January	2,437	Peak shifts to winter
2024	October	2,474	Peak shifts to autumn

The peak lands in a different month each year (August, January, October), and the top four months overall are within 1.2% of each other. In practice no month is a reliable peak: the highs are only about 5% above average and do not repeat, so a rolling forecast around a flat baseline fits better than a fixed seasonal plan.

Quarter-wise Shipment Trend

Quarter	Loads	Change vs Prior Quarter	% Change
Q1-2022	7,015	—	—
Q2-2022	7,150	+135	+1.9%
Q3-2022	7,221	+71	+1.0%
Q4-2022	7,203	−18	−0.2%
Q1-2023	6,985	−218	−3.0%
Q2-2023	6,996	+11	+0.2%
Q3-2023	7,104	+108	+1.5%
Q4-2023	7,080	−24	−0.3%
Q1-2024	7,099	+19	+0.3%
Q2-2024	7,214	+115	+1.6%
Q3-2024	7,159	−55	−0.8%
Q4-2024	7,184	+25	+0.3%

Quarterly volume stays between 6,985 and 7,221 loads, a spread of about 3%. The biggest move is Q1-2023 (−218 loads, −3.0%); every other change is within ±2%. Q1 is the lowest quarter each year, mainly because it has fewer days. There is no sustained growth or decline.

Month-over-Month Load Growth

Of 35 monthly changes, 16 were increases and 19 decreases. The big swings are the February drop and March rebound (February 2023 −301, −12.4%; March 2023 +276, +12.9%), plus September 2022 (−192) followed by October 2022 (+164). Otherwise changes are mostly within ±100 loads and reverse the next

month. From January 2022 to December 2024 volume is down 113 loads (−4.6%), but January 2022 was one of the highest months, so that is not a real decline.

Revenue Contribution by Season

Season	Months	Seasonal Revenue	Share	Average per Month
Monsoon/Rainy	Jul – Sep	$65,801,076.40	25.06%	$7,311,231
Winter	Dec – Feb	$64,369,879.62	24.52%	$7,152,209
Summer/Pre-monsoon	May – Jun	$44,424,554.67	16.92%	$7,404,092
Autumn/Post-monsoon	Oct – Nov	$43,996,926.76	16.76%	$7,332,821
Spring	Mar – Apr	$43,933,362.84	16.73%	$7,322,227
Total	**12 months**	**$262,525,800.29**	**100%**	**$7,292,383**

The shares mostly reflect how many months each season covers: Winter and Monsoon/Rainy have three months each (about 25%) and the other three have two (about 17%). Per month, seasons range from $7.15 million (Winter) to $7.40 million (Summer/Pre-monsoon), about 3.4% apart, with Winter lowest because it includes February. Revenue is spread evenly through the year, so seasonal pricing power looks limited.

> **Key Question — Resolved:** *Which month had the highest load volume? August 2022 recorded the highest single-month volume of 2,496 loads, with no tie (next highest: October 2024 at 2,474). It is about 5% above the monthly average and should be read as the highest observed month, not as a recurring August peak.*

Data Sources

Tables: loads, trips

Join Path: Time-based aggregation using load_date and dispatch_date

14. Cross-Module Key Findings

The following findings draw together the most significant insights from the eight modules to inform cross-functional decision-making.

Driver & Delivery Performance

- The fleet on-time rate is 55.68% and every driver sits between 51.61% and 58.98%, so late deliveries (about 44% of events) appear to be a systemic challenge, not a driver-specific one. A fleet-wide process review is likely to help more than individual coaching, with DRV00133 and DRV00074 (about 4 points below fleet) the only clear individual outliers.
- DRV00101 leads on revenue per mile ($2.21), but the whole driver base falls within about 6% ($2.09 to $2.21), and DRV00101 itself is below the fleet on-time rate (54.14%). Preferential load assignment to the top cohort is a modest lever, not a large one.
- Driver MPG (6.45 to 6.56) is almost identical across the fleet, so fuel efficiency does not separate drivers.

Route Economics & Fuel Management

- RTE00016 is the most profitable lane after fuel ($8.46M) and also the highest-revenue lane in December 2024, so it is the lane to protect with capacity and contract security.
- Seven routes lose money after fuel (RTE00010 the worst at −$1.42M). With fuel cost per route almost constant, the difference is on the revenue side, which points to pricing and load mix.
- Fuel cost is rising on seven routes, most sharply on RTE00028 (+95% from October to December 2024).
- Fuel cost per mile ($0.39 to $13.69) is not consistent with near-identical route MPG and needs data validation before use.

Fleet Asset Management

- TRK00003 combines the highest maintenance cost ($90,161.42) with the highest downtime (1,133.1 hours), a concentrated asset-health risk.
- TRK00062 has the highest downtime in the last six months (329.9 hours) and should be reviewed before further downtime accumulates.
- Mileage is fairly even across 92 trucks (the highest-to-lowest gap equals about 18% of the average), with 44 below the average. TRK00040 is the clearest case for reassignment.
- Maintenance cost is a small share of truck revenue (about 0.4% to 2.7%), and downtime shows no clear effect on trip volume.

Customer Portfolio

- Revenue is highly diversified: the largest customer holds 0.59% and the top ten 5.7%. Concentration risk is low, with correspondingly limited leverage from any single account.
- Revenue per load varies by about 22% from lowest (CUST00149, $2,695.11) to highest (CUST00070, $3,294.30), which makes yield, not volume, the main commercial lever.
- The top customer changes every quarter and month-level growth is mean-reverting, so single-period rankings should not drive account decisions.

Safety & Compliance

- Risk is concentrated: DRV00058 and DRV00003 together account for 13 incidents, and 18 drivers with three or more incidents account for about 36% of driver-linked incidents.
- 37.6% of incidents (64 of 170) were preventable, which is the portion that behaviour and training programmes can address. DRV00070, DRV00098 and DRV00104 had all three of their incidents classed as preventable.
- Incidents rose from 54 to 60 in 2024, but at these volumes the change is within normal variation and warrants monitoring.

Demand Planning

- Demand is flat (about 2,370 loads and $7.29 million per month) with no repeatable peak season. The monthly peak moved from August 2022 to January 2023 to October 2024.
- February is consistently the lowest month, mostly because it is shorter, so month-to-month comparisons should be normalised by days.
- Revenue is spread evenly across seasons (about $7.15M to $7.40M per month), so capacity planning can use a steady baseline with rolling monthly review.

15. Strategic Recommendations

These recommendations come straight from the findings. Each has a priority, an owner and the impact we expect.

Priority	Recommendation	Owner	Module	Expected Impact
HIGH	Run a focused safety review for DRV00058 and DRV00003, and coach DRV00070, DRV00098 and DRV00104, whose incidents were all preventable. Keep watching the 2024 incident level.	Compliance & Safety	7	Fewer incidents; lower insurance and regulatory exposure
HIGH	Review the seven loss-making routes (RTE00010 first) on pricing, load density and distance before adding more volume.	Finance / Operations	2	Stop negative-margin lanes
HIGH	Look into rising fuel cost on RTE00028 (+95%) and the six other routes that rose three months running.	Finance / Operations	2	Stop fuel cost creep
HIGH	Assess TRK00003 for an overhaul or replacement, given its cost and downtime.	Fleet Management	4	Less unplanned downtime
MEDIUM	Schedule a maintenance review for TRK00062 (329.9 hours of recent downtime) and check TRK00014's cost per mile.	Fleet Management	4	Avoid service disruption
MEDIUM	Move work to trucks below average mileage, starting with TRK00040.	Fleet Management	3	Better fleet utilisation
MEDIUM	Confirm how fuel purchases are tied to routes before relying on fuel cost per mile, then revisit RTE00015 and RTE00025.	Planning / Data	5	Reliable fuel metrics
MEDIUM	Review rates for the lowest revenue-per-load accounts (CUST00149, CUST00025, CUST00148) and plan growth for CUST00070.	Sales	6	Higher yield per load
MEDIUM	Look at the fleet-wide on-time rate (55.68%) through scheduling and appointment windows, and follow up with DRV00133 and DRV00074.	Operations	1	More reliable delivery
LOW	Set up route-level MPG monitoring and check RTE00021, the only route with a sustained decline.	Transportation Planning	5	Catch efficiency loss early
LOW	Plan capacity around a flat baseline of about 2,370 loads a month, and compare months per day, not per month.	Strategic Planning	8	More accurate capacity plans

16. Conclusion

This report presents the findings of the Logistics Operations Analytics project. Across eight modules, the Business Analytics Team turned raw Transportation Management System data into evidence that leadership can use for operational and strategic decisions.

The analysis shows an operation that is stable and evenly balanced. Demand is flat across months and years, customer revenue is widely spread, and driver and route efficiency metrics sit in narrow bands. The meaningful differences are concentrated in a small number of places: a few loss-making and rapidly rising-fuel-cost routes, a single truck (TRK00003) carrying the highest maintenance and downtime burden, and two drivers (DRV00058 and DRV00003) at the top of the incident rankings.

The most urgent actions are the safety review of the highest-incident drivers, the review of loss-making and rising-cost routes, and the asset assessment of TRK00003. Medium-term priorities include fleet rebalancing, customer yield review and validation of the fuel cost per mile metric. Longer term, the organisation should consider predictive maintenance scheduling and route-level fuel monitoring as embedded capabilities.

It is recommended that the report be reviewed in a cross-functional leadership session, that named owners be assigned to each High and Medium priority recommendation, and that a 90-day review measure the impact of the actions taken. The Business Analytics Team can support deeper analysis of any module, develop dashboards from these findings or assist with implementation planning.

— End of Report —

Business Analytics Team | Third-Party Logistics Company | October 10, 2026